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Project Management 4.1

Project Risk Management

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Learning Outcomes

- By the end of this week, you will be able to:
  - Compile a
    - risk register
    - risk plan
  - State why risk management is used in projects and the wider environment
  - Discuss various methods to manage risks
  - Identify project risks
  - Give relevant risk examples and how to mitigate them.

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Definition

- Risk is defined as “an uncertain event or set of events which, should it occur, will have an effect on the achievement of objectives”

A risk statement begins with

- “There is a risk that...”

Source: OGC M_o_R 2007

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How Do You Perceive Risk?

- Spend a few minutes discussing risk
  - How do you perceive risk?
  - Have you any experience of risk management?
  - Prepare to offer views

Risk

- We manage risks every day throughout our lives
  - When we play sport
  - When we ask someone out on a date or play the lottery
  - When we travel
  - When we shop on-line
  - When we choose a career
- We are all constantly evaluating risks, prioritising them and mitigating them

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Risk

- Risk when we travel:
  - There is a risk that I will be delayed and be late for my appointment
  - I evaluate that risk according to my experience or perception of the reliability of the mode of transport I intend to use
  - I mitigate the risk by the time I start my journey relative to the time of my appointment.

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- Risks
  - *“Project Risk: An uncertain event or condition that, if it occurs, has a positive or negative effect on a project objective”*
    - **Risk = FUTURE (may occur)**
- Issue
  - An event that requires a decision otherwise a negative effect on the project may result.
    - **Issue = NOW (has occurred)**

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Projects and Risk

- Can there ever be such a thing as a risk free project?
  - There is a risk
  - Projects will have to manage risks
  - Risk is considered at every step
  - There is **always** risk.

Good and Bad

- A combination of:
  - **PROBABILITY** of a perceived **THREAT** or **OPPORTUNITY** occurring and
  - Magnitude of its **IMPACT** on objectives
- A threat can be seen as a **negative** impact on the objectives
- An opportunity can be seen as **favourable** impact on the objectives.



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Aims of Risk Management

- Support improved decision making
- Understand risks and their likely importance or the risk exposure
  - Probability of occurrence
  - Impact if threat/opportunity occurs
  - Proximity of threat/opportunity
- Prioritise what we focus on with our limited resources (time, cost etc).

Risk Management

- Systematic application of:
  - Principals
  - Approach
  - Processes to the tasks of:
    - Identifying
    - Estimating
    - Evaluating
    - Planning
    - Implementing Risk Responses.



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Attitude aka Risk Tolerance

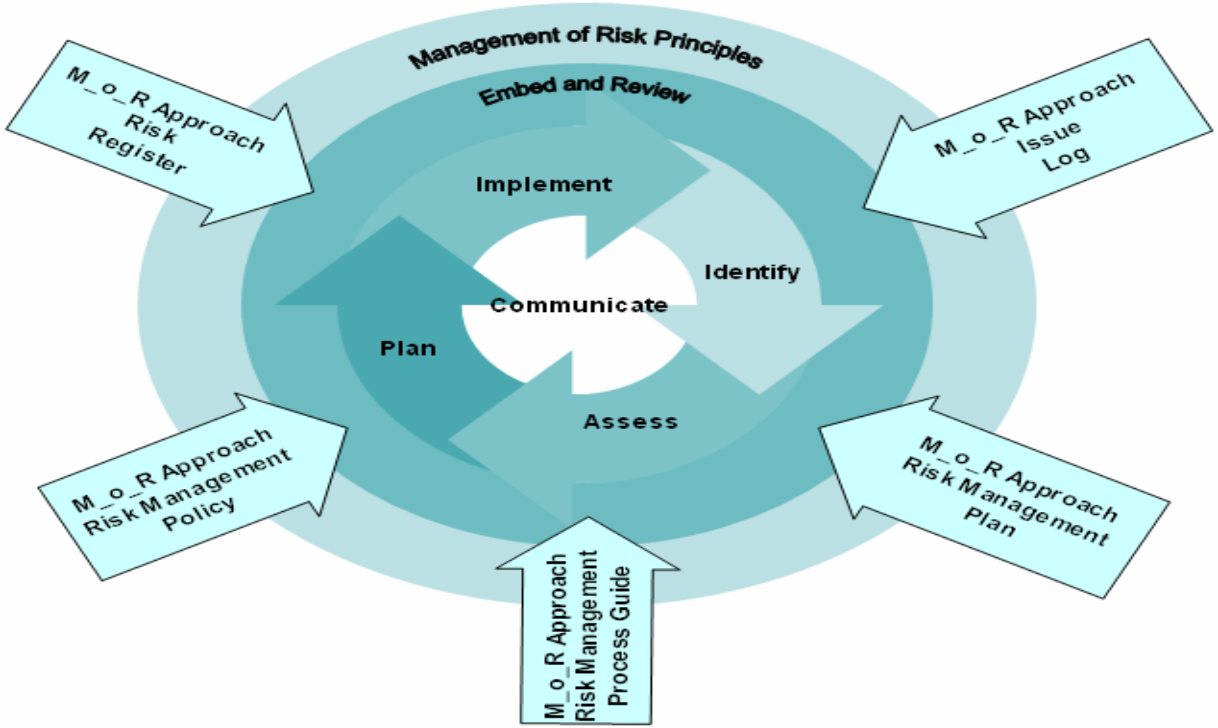
- Your attitude to risk will affect how you perceive the risks you face compared to someone else
- A systematic approach can help compensate for that bias to ensure risks identified by different people can be compared and prioritised.
- There is a difference between risk tolerance in an individual, and risk tolerance in an organisation.

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PRiNCE2 Management of Risk Framework

One view of Risk:

MoR Model



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Benefits of Managing Risk

- Corporate decision making
- A progressive management style
- Objectives realised
- Corporate governance met
- Ownership and accountability
- Financial benefit
- Greater visibility
- Consistency
- Environment
- Improved contingency plans
- Improved awareness

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Managing Risk in 4 Steps

1. Risks are Identified
2. Risks are Analyzed and Prioritized
3. Actions are taken to minimize the effect of risk
4. Actions are monitored to ensure continued effectiveness

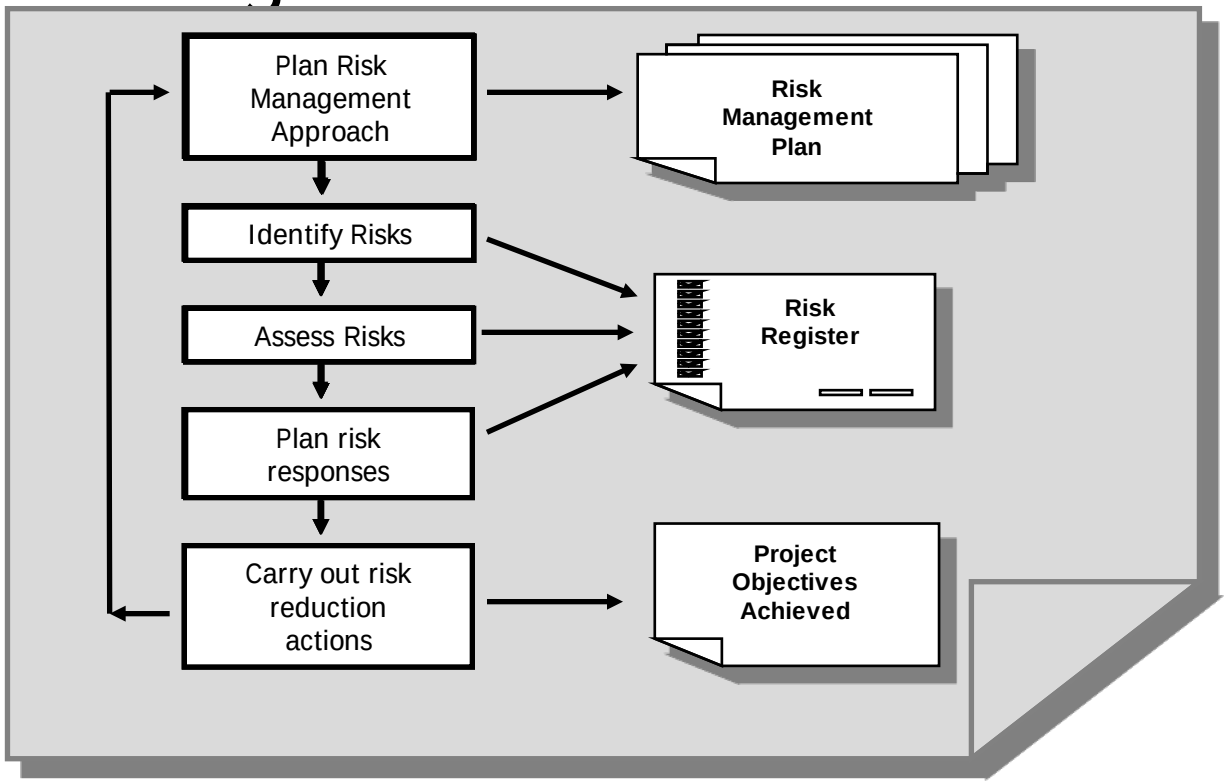


Microsoft Word Document

Risk Management Process Template

- Risk Management should be present at all stages of the project lifecycle
- Risk Management aims to assess the likelihood of a risk issue occurring and to determine the impact of the risk issue on the project
- Time spent on risk management as part of the project management process will improve the likelihood of project success.

The Risk Management Process



Cadle and Yeates advocate a similar process.

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Risk Appetite (Risk Tolerance)

Probability		Very Low	Low	Medium	High	Very High
		0.05	0.1	0.2	0.4	0.8
0.9	Very High 71-90%	0.045	0.09	0.18	0.36	0.72
0.7	High 51-70%	0.035	0.07	0.14	0.28	0.56
0.5	Medium 31-50%	0.025	0.05	0.10	0.20	0.40
0.3	Low 11-30%	0.015	0.03	0.06	0.12	0.24
0.1	Very Low up to 10%	0.005	0.01	0.02	0.04	0.08

Impact

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Risk Responses

Avoid

Mitigate

Transfer

Accept

Share

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Risk Management Response Plan

- **What is a Risk Response Plan?**
  - A Risk Plan outlines the foreseeable project risks and provides a set of actions to be taken to both prevent the risk from occurring and reduce the impact of the risk should it occur.

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Plan Contents



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Risk Plan Template

- A full list of all of the foreseeable risks during the project – is this practical?
- A rating of the likelihood of each risk's occurring
- A rating of the impact on the project should each risk actually occur
- A priority rating of the overall importance of each risk
- A set of preventative actions to reduce the likelihood of the risk's occurring
- A set of contingent actions to reduce the impact should the risk eventuate
- A process for managing risks through the project.

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Risk and Issue Registers

- Risk Register
- The Risk Management Strategy
- Every project has risks.

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Risk & Issue

- Each risk on the Risk Register is allocated a unique identifier as well as details such as:
  - Who raised the risk
  - When it was raised
  - The category of risk
  - The description of the risk (cause, risk event, effect)
  - Probability, impact and expected value
  - Proximity
  - Risk response category
  - Risk response actions
  - Risk status
  - Risk owner
  - Risk actionee

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Risk & Issue Register

Risk Log

Reference	Description	Date	Likelihood	Impact	Risk Category	Countermeasure	Contingency	Owner
1	The main supplier cannot deliver on time because of other commercial commitments.	21/03/11	Likely	High	High	Include financial penalties in contract; build contingency into the schedule; monitor contractor performance.	Revise project schedule.	Annie Proulx
2	The lead time for the leased line exceeds 90 days.	21/03/11	Unlikely	Medium	Medium	Order line earlier than necessary; incur additional rental fees.	-	Jim Harrison
3	Release of the new system is delayed because user acceptance testing commences after planned start.	23/03/11	Very likely	High	High*	Build eight week contingency into schedule.	Employ temporary staff to free up resources for testing; revise project schedule.	Mark Spragg
4	There is insufficient capacity to create additional database instances for data migration and testing.	18/04/11	Very unlikely	Medium	Low	Give advanced notice; monitor SAN usage.	Prioritise projects; temporarily remove alternative development instance.	Jim Harrison
5	Load testing fails to identify problem prior to system release.	20/04/11	Unlikely	High	Medium	Phased introduction of new system users.	Manage system access; remove high-load features; initiate incident management process.	Kent Haruf

Issue Log

Reference	Description	Date	Raised by	Owner	Impact/ Priority	Risk	Progress/ Resolution
1	User acceptance testing will begin after scheduled start.	27/07/11	Mark Spragg	John Steinbeck	Medium	3	Backfill permanent posts with agency staff so they are available to start testing week commencing 13/08/11. Revised project schedule (contained within contingency.) Continue to monitor situation.
2	Interface with document management system is very slow at certain times.	28/07/11	Harper Lee	Kent Haruf	Medium	6	Under investigation. Report due 01/08/11.
3	Conference room pilot users report data loss – documents not registered with document management system.	01/08/11	Mark Twain	Kent Haruf	High	9	Escalate to board and suppliers.
4	Server log file fills partition after a few days.	01/08/11	Jack Kerouac	Jim Harrison	Low	10	Regularly truncate log file. Raise issue with supplier.
5	No time was allocated to train new recruits.	01/08/11	Willa Cather	Mark Spragg	Medium	13	

http://martinwebster.eu/2011/04/22/how-to-prepare-an-issue-log/

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Successful Risk Management

- Consider the Organisational/Human issues
- Develop a risk management culture
- Emphasis on positive thinking
- Learn from previous mistakes
- Team members that raise awareness of risks to the project are thanked
- Encourage the process of reflecting upon and reviewing each project
- Someone should be responsible for raising the awareness of risk management
- One member of the team responsible for ensuring that the risk management procedures are undertaken
- Ensure that risk management is carried out throughout the life cycle of the project
- Reporting should never be seen as ‘going through the motions’
- Allow the project manager to be pro-active.

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Risk Management

- Risk analysis should not paralyse a project and stop it progressing
- On the contrary it should help the project assess the way forward and help identify possible problems so they can be avoided
- Use risk reporting to get the help and support you need to move your project forward and to assist stakeholders to understand the risks the project faces.